

RESERVOIR STAGE ANALYSIS – LAB 02

Fitted Model and Parameters

The reservoir stage data were modeled using a **two-logistic model** to represent the rising and falling stages of the reservoir. The fitted parameters were highly significant, with all **p-values < 0.001**, indicating that the model coefficients are statistically well-determined. The baseline stage was **14.3493 m**, while the rise amplitude and rate were **8.9160 m** and **0.5117/h**, respectively. The rise midpoint occurred at **31.5198 h**. For the falling stage, the amplitude was **-3.7212 m**, with a rate of **0.2831/h** and a midpoint of **37.5461 h**.

Model Performance

The model produced an excellent overall fit. The **Sum of Squared Errors (SSE)** was **1.801659 m²**, while the **coefficient of determination (R²)** was **0.999114**, meaning that approximately **99.9% of the variation** in reservoir stage was explained by the model. The residual standard error was only **0.080072 m**, or about **8 cm**. The calculated **Area Under the Curve (AUC)** was **1260.970821 m·h**, corresponding to a mean stage of **17.5745 m** over the **71.75-hour** monitoring period.

Residual Analysis

The residuals were generally small and centered near zero, with a maximum absolute residual of **0.2208 m**. More than **95% of the residuals** were within ± 2 standard errors. However, the residuals showed a strong pattern, with only **17 sign runs compared with 192 expected**, and a longest consecutive same-sign run of **54 points**. This indicates that the remaining errors are systematic rather than purely random.

Conclusion

The two-logistic model provides a **high-quality and statistically significant representation** of reservoir stage behavior. Its **99.9% variance explained** and approximately **8-cm typical error** make it suitable for estimating cumulative volumes, mean water levels, timing, and overall hydrograph characteristics. However, the patterned residuals indicate that the model may not accurately capture subtle changes in stage behavior, such as asymmetric rise and fall or stage-dependent storage effects. For applications requiring centimeter-level instantaneous accuracy, a more flexible model with additional terms or time-varying parameters may provide better predictions.